

JOSE ANTONIO ELIZALDE

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Education

University of California, Los Angeles

Sept 2025 – June 2027

B.S. in Computer Science and Engineering

Westwood, CA

Experience

UCLA ACM Studio | Jump The Gun | *Unity, C#, Git* – Game Director

Oct 2025 – Present

- Led the design and development of *Jump the Gun*, a first-person action shooter, defining gameplay mechanics and coordinating implementation across an **8-person team** of programmers, artists and designers.
- Implemented a custom **physics-based movement controller** in **C#**, replacing Unity's default character controller with manually programmed movement and collision logic.
- Maintained the team's **Git branching and pull-request workflow**, reviewing and integrating gameplay features developed concurrently by multiple contributors.

Projects

DailyCodeForce | *Next.js, TypeScript, PostgreSQL, Prisma, Go, Docker, Nginx* Aug 2026 – Present

- Architected a self-hosted competitive programming platform orchestrating **7 Docker Compose services** – Next.js web app, PostgreSQL, sandboxed judge, Go Discord bot, presence tracker, cron scheduler, and **Nginx reverse proxy** – deployed on **DigitalOcean**.
- Built a **sandboxed C++17 judge** executing untrusted submissions in **under 1s** under **four enforced rlimits** (256MB address space, 3s CPU, 64 processes, 10MB writes) inside a hardened container with pids caps, a noexec /tmp mount, and no-new-privileges.
- Implemented scheduling with a **90-day deduplication window** and **30-day automated backfill** over a **113-problem** PostgreSQL pool managed via **Prisma**, delivering **81 consecutive daily problems with zero repeats**.

Robinhood Discord Portfolio Bot | *C++, Python, FastAPI, pandas, NumPy*

June 2026 – Present

- Engineered a **C++** Discord bot (D++) and Python **FastAPI** backend serving real-time Robinhood portfolio analytics at **486ms median latency** and **783ms p99** across 200 sampled requests, running on a **single-vCPU 512MB** Linux VPS.
- Architected a **decoupled two-service system** with pickled session persistence and proactive token refresh, restoring authenticated brokerage sessions across restarts without re-login and reaching full availability **2s after cold start**.
- Built a two-stage screening pipeline evaluating all **503 S&P 500 tickers** on momentum in **39s** before applying fundamental filters, running as a **background thread** within a **107MB peak memory footprint** under **systemd**.

Personal Developer Portfolio | *React, Next.js, Tailwind CSS*

Aug 2026 – Sept 2026

- Engineered a responsive, high-performance developer portfolio to showcase projects, technical skills, and research.
- Architected the frontend UI/UX using **React** and **Next.js**, optimizing for cross-device accessibility, **Lighthouse** performance metrics, and SEO.
- Configured a **CI/CD** workflow connected to GitHub so pushes to main build and publish the site automatically.

Shellscape | *Gameplay Programmer* | *C#, Unity, WebGL*

Aug 2026

- Shipped a browser-playable action game with a **team of five** for Club Club Jam Jam 2026, designed and completed inside a game jam window, placing **74th out of roughly 150 teams** and released publicly on itch.io.
- Implemented the **movement and interaction systems**, including the body-possession mechanic the game is built around, and built the tutorial scene with its scripted narration and dialogue boxes.

Research

Carnegie Mellon University – REUSE Research Program

May 2025 – Aug 2025

Undergraduate Researcher, Human-Computer Interaction Institute (HCII)

Pittsburgh, PA

- Co-authored a paper with CMU faculty that was accepted to **CHI 2026** (ACM Conference on Human Factors in Computing Systems).

University of California, Los Angeles – SURE C² Research Program

Aug 2024 – May 2025

Undergraduate Researcher, Department of Mathematics

Los Angeles, CA

- Researched transfer systems in combinatorics on partially ordered sets, discovering a recursive pattern on X_n^{++} and co-authoring a paper presented at UT Arlington.
- Developed a **C++** program to automate the counting of all transfer systems on the poset X_n^{++} .

Technical Skills

Languages: C++, C, C#, Python, Go, TypeScript, SQL, Bash

Backend & Infrastructure: Linux, Docker Compose, FastAPI, PostgreSQL, Prisma, Nginx, systemd, DigitalOcean

Web & Data: Next.js, React, Tailwind CSS, pandas, yfinance

Tools: Git/GitHub, CMake, Unity, Vim/NeoVim, LaTeX